

Kids First Data Resource

Vision and DRC Participants

Gabriella Miller Kids First Vision: Alleviate **suffering** from childhood cancer *and* structural birth defects by fostering **collaborative research** to uncover the etiology of these diseases and supporting **data sharing** within the pediatric research community.

DRC Funder: NIH Common Fund's Gabriella Miller Kids First Pediatric Research Program; administered by NHLBI (PO: Charlene Schramm) overseen by [NIH Kids First Working Group](#).

DRC PIs: Adam Resnick, Brandi Davis-Dusenbery, Vincent Ferretti, Robert Grossman, Allison Heath, Deanne Taylor, Sam Volchenbom

DRC Institutions: Children's Hospital of Philadelphia, CHU Sainte-Justine, University of Chicago, Seven Bridges Genomics

Go Live Date: September 10, 2018

Data Available

Portal: <https://portal.kidsfirstdrc.org/>

Kids First Released Datasets (22K + Participants):

- 21 Kids First datasets released as of 9/14 (more in coming weeks)
 - **14 Structural Birth Defects datasets** spanning congenital diaphragmatic hernia, congenital heart defects, orofacial cleft, other craniofacial anomalies, disorders of sex development etc.
 - **7 Childhood Cancer datasets** spanning neuroblastoma, ewing sarcoma, leukemia, enchondromatoses etc
 - **1 cohort of children with both structural birth defects and cancer**
- + 5 fully interoperable datasets (e.g., TARGET, CBTN)

Total 44 datasets, 48,000 genomes, 20,000 cases approved for Sequencing as Part of Kids First ([Link](#))

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Interoperability Use Cases

We have a number of researchers that will be working across datasets in multiple platforms. In general, pediatrics and developmental biology spans organs, disease and data types - making effective and sustained interoperability a critical need.

Examples of overlapping cohorts:

INCLUDE: KFDRC and INCLUDE Data Hub

PCGC: KFDRC and BDC

CMG: KFDRC and AnVIL

CSER: KFDRC and AnVIL

NBL: KFDRC and NCI CRDC

AML: KFDRC and NCI CRDC

NCBI: KFDRC and UDN

Brain Tumors/CCDI: KFDRC and NCI CRDC

Early on, we realized we needed “intra”-operability for releases and developed a release coordination protocol. Perhaps an area for interoperability as well.

Clinical/Phenotypic Data Interop Pilot Opportunities

Clinical and phenotypic data has been identified as a limiting factor across supported studies. The wide range provides challenges, but also opportunities for interoperability. Areas under investigation include:

- FHIR as a core standard
- Terminologies for harmonized and computable data values
- Identifying virtual cohorts across the platforms based on clinical and phenotypic data
- Workflows for clinical/phenotypic data harmonization

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Tools Available

Kids First DRC Workflows - functional equivalency to other large datasets/resources

- [Germline](#): Trio/family-based *GATK* germline best practices
- [Somatic](#): *Strelka2*, *Mutect2*, *Lancet*, *VarDict*, *Manta*, *Control-FREEC*
- [RNA-Seq](#): *STAR 2-pass*, *RSEM*, *Kallisto*, *STAR-fusion*, *Arriba*

CAVATICA - data analysis and sharing platform

- Over 450 public apps
- Interactive analysis with Jupyter Notebooks and RStudio

Other Tools

- **Variant Search** - explore annotated Kids First variants across datasets in your browser
- **PedCBioPortal** - open access visualization and analysis of somatic mutations and expression
- **Variant Workbench** - query Kids First data within a Zeppelin environment using SQL and PySpark (*coming soon*)

Authentication, Authorization and Indexing

Authentication:

- **Controlled access:** eRA Commons
- **Registered access:** Google, Facebook and ORCID (OAuth2)

Authorization:

- **NIH Datasets:**
 - Telemetry reports from dbGaP either to Kids First Framework Services or NCI CRDC Framework Services
- **Consortium Datasets:**
 - Whitelist via Gen3 (also eRA Commons AuthN)

Indexing:

- The Kids First Data Service integrates with Gen3 indexd to index and associate relevant data for querying to the files. Cavatica also recently released GA4GH DRS endpoints for all files.

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Architectural Diagram - Technical

